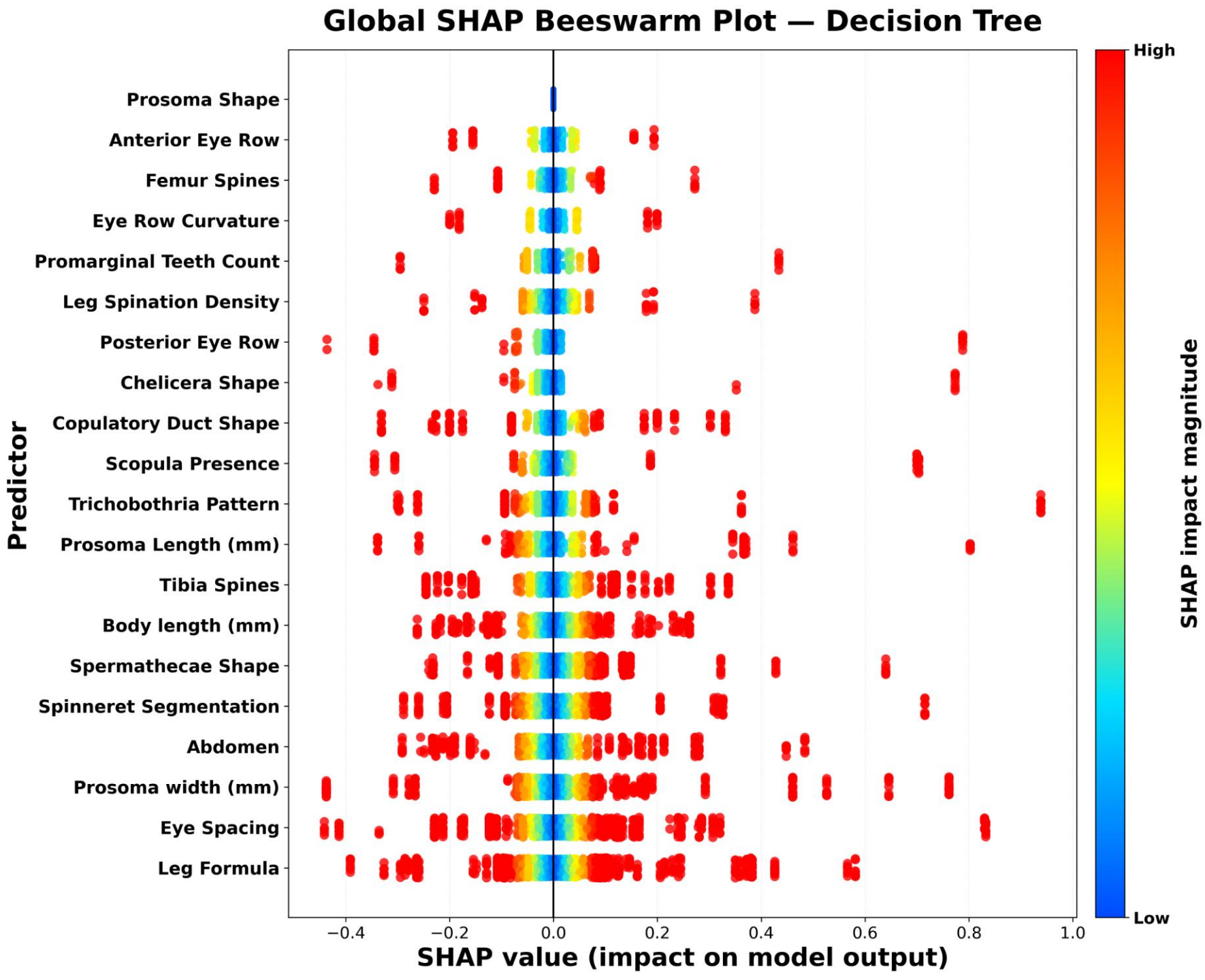
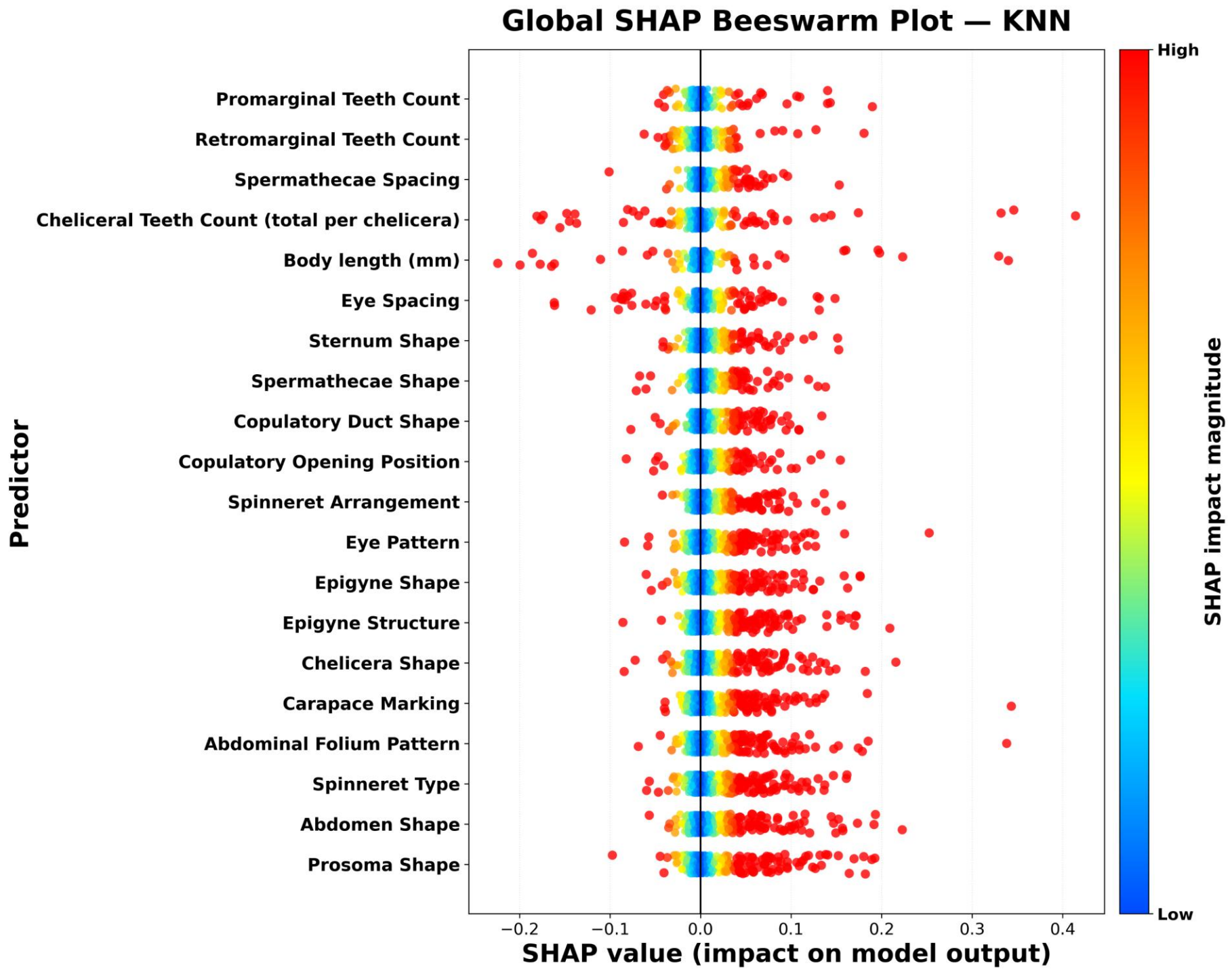
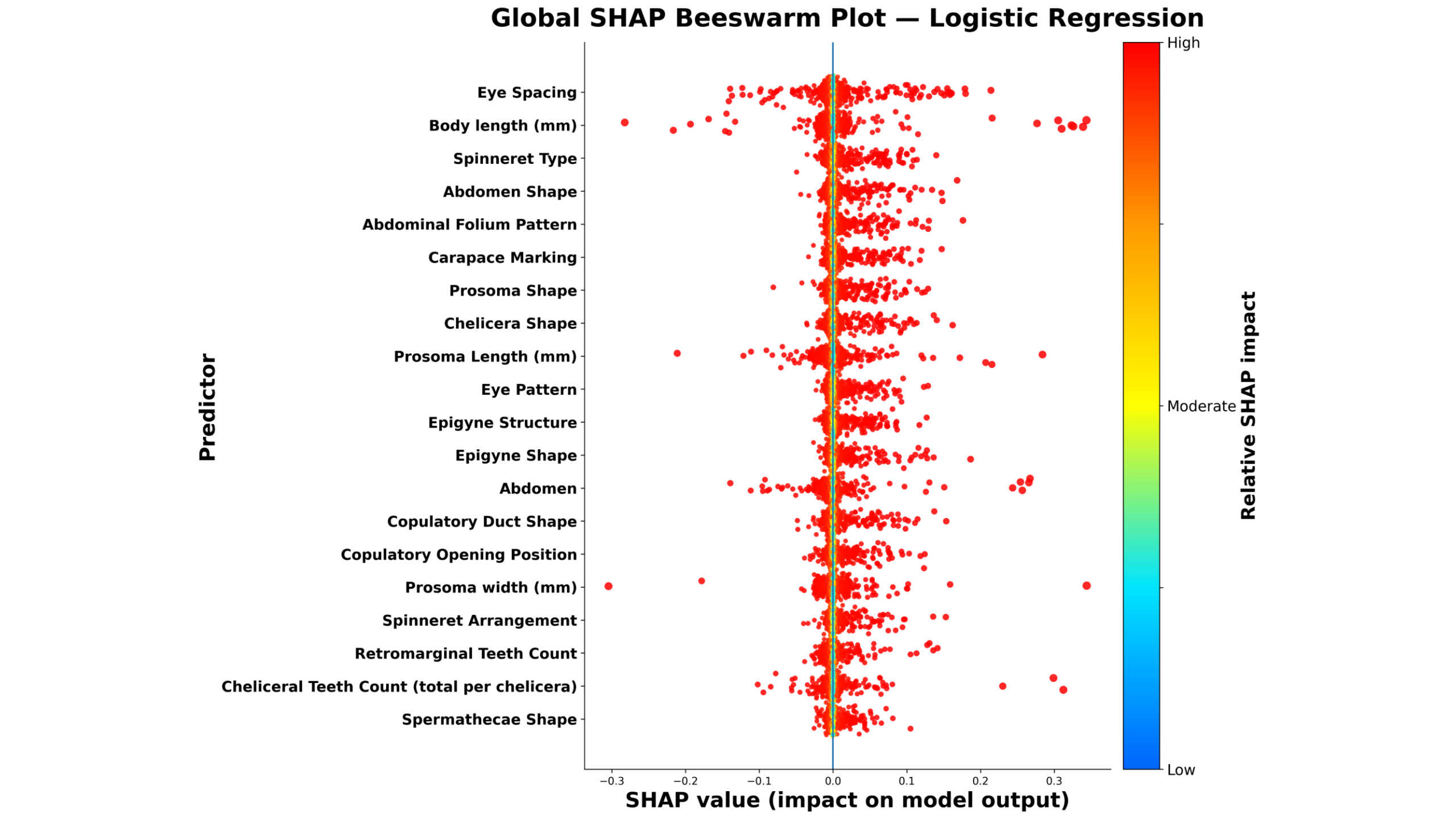
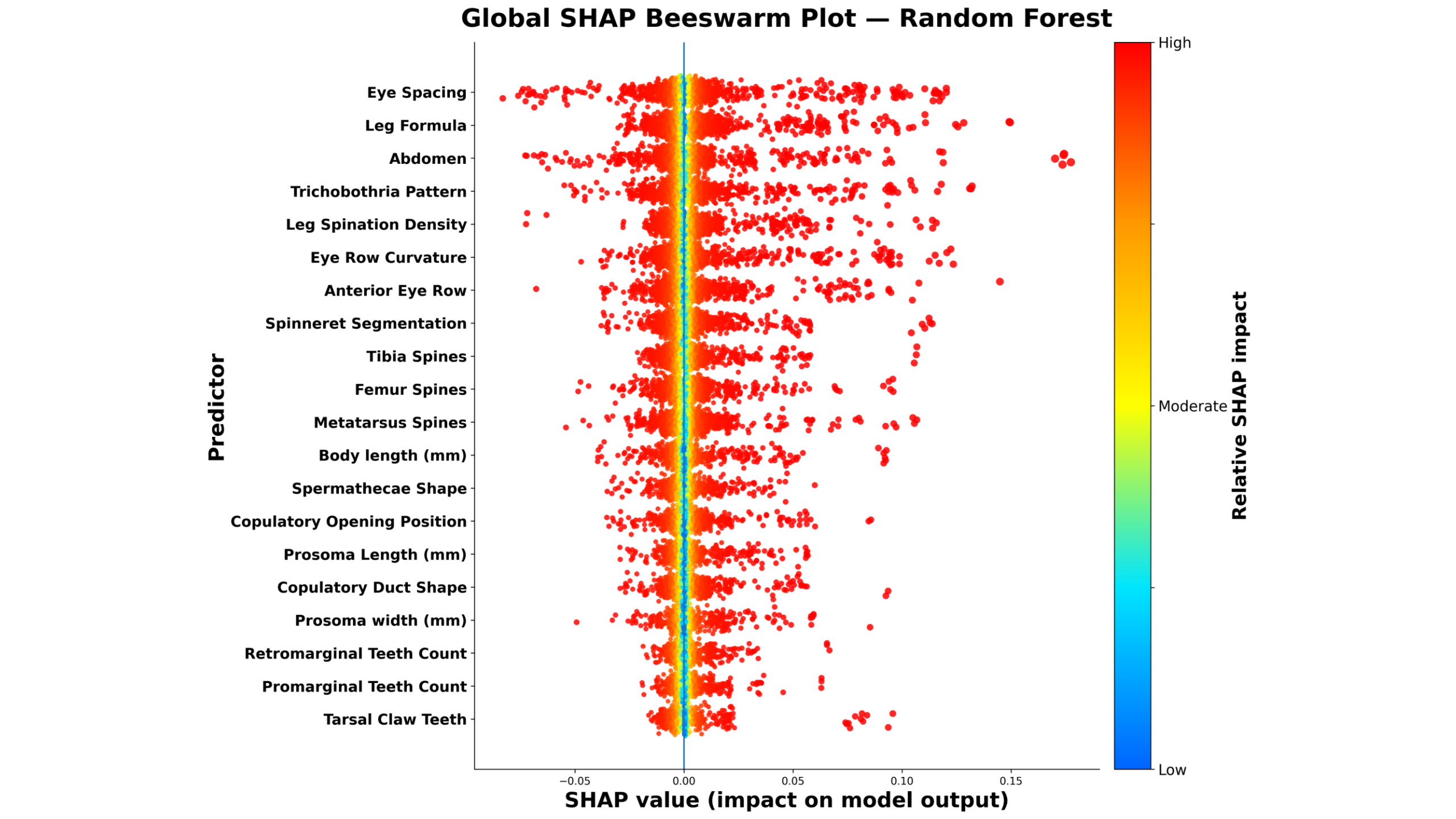
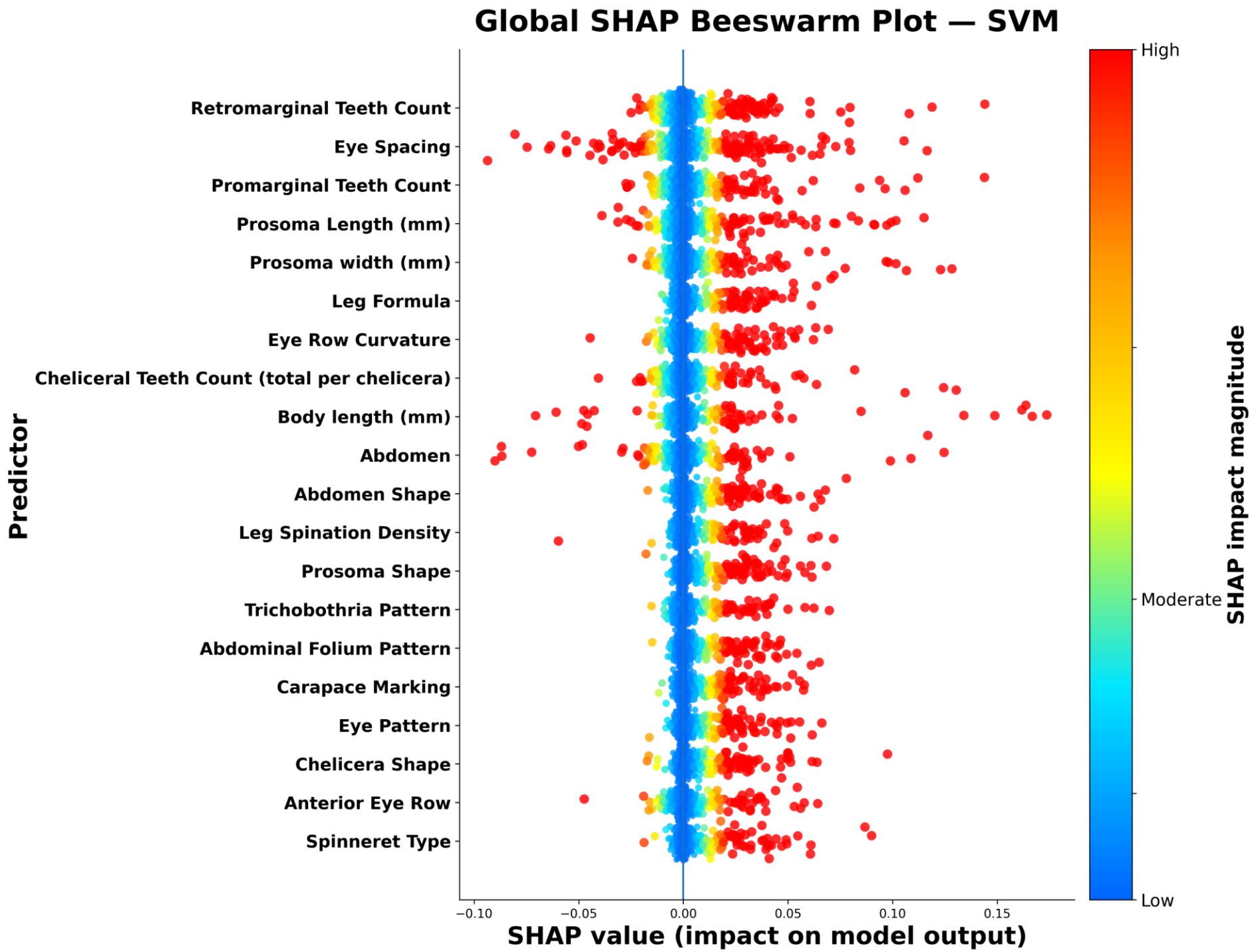
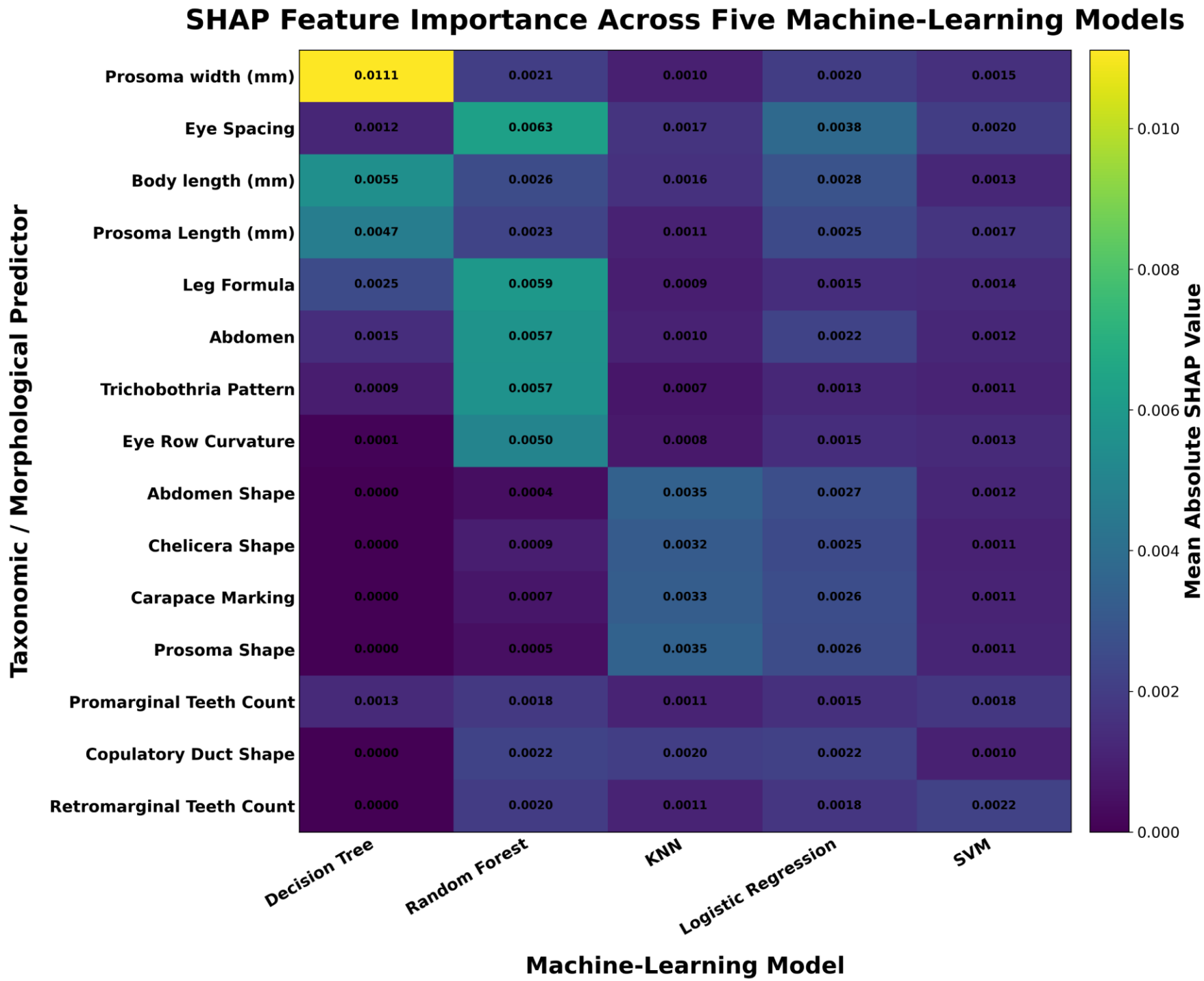

Global SHAP Beeswarm Plot — Decision Tree
Predictor
Prosoma Shape
Anterior Eye Row
Femur Spines
Eye Row Curvature
Promarginal Teeth Count
Leg Spination Density
Posterior Eye Row
Chelicera Shape
Copulatory Duct Shape
Scopula Presence
Trichobothria Pattern
Prosoma Length (mm)
Tibia Spines
Body length (mm)
Spermathecae Shape
Spinneret Segmentation
Abdomen
Prosoma width (mm)
Eye Spacing
Leg Formula
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
SHAP value (impact on model output)
High
Low
SHAP impact magnitude

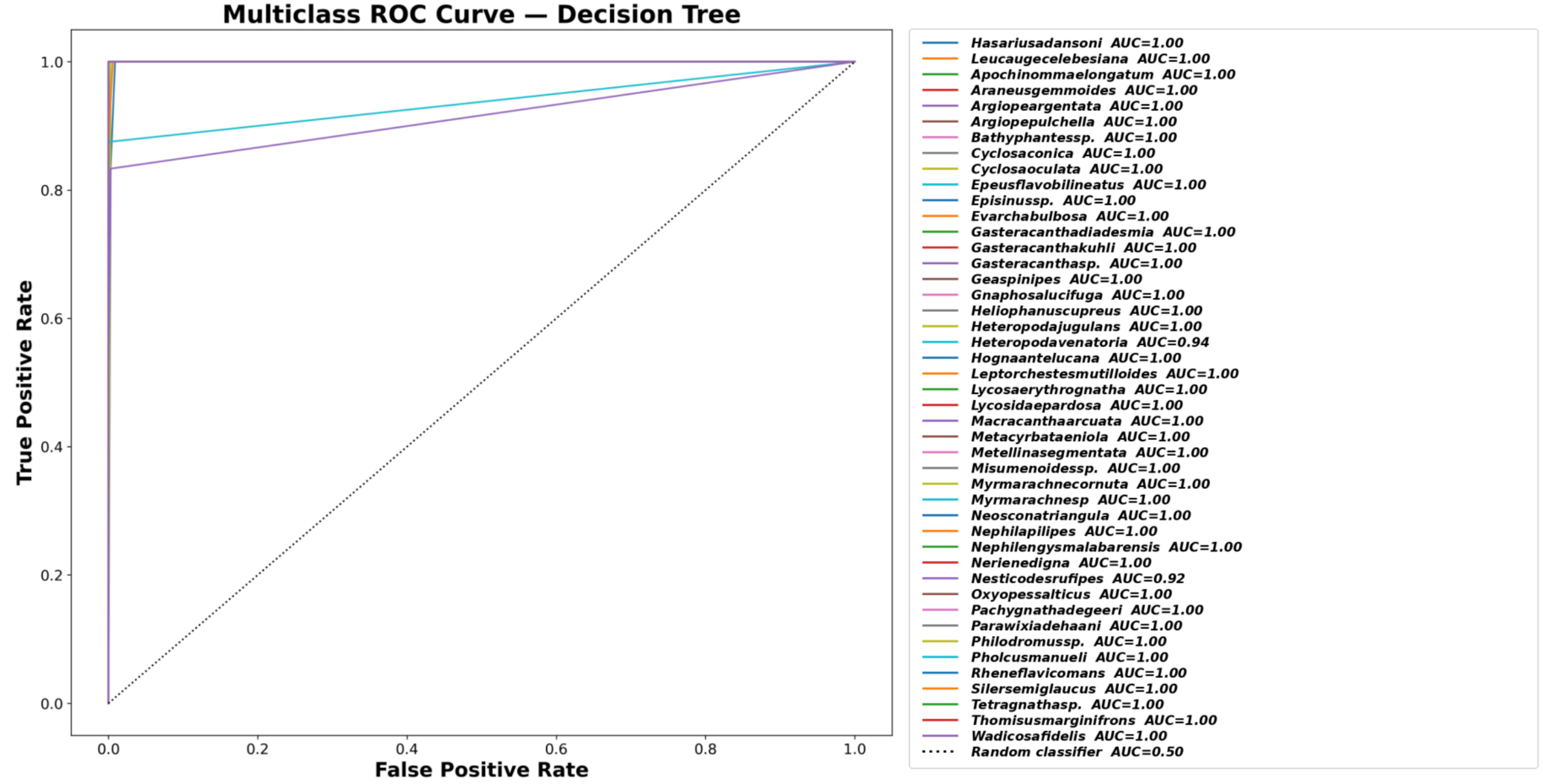

Global SHAP Beeswarm Plot — KNN
Predictor
Promarginal Teeth Count
Retromarginal Teeth Count
Spermathecae Spacing
Cheliceral Teeth Count (total per chelicera)
Body length (mm)
Eye Spacing
Sternum Shape
Spermathecae Shape
Copulatory Duct Shape
Copulatory Opening Position
Spinneret Arrangement
Eye Pattern
Epigyne Shape
Epigyne Structure
Chelicera Shape
Carapace Marking
Abdominal Folium Pattern
Spinneret Type
Abdomen Shape
Prosoma Shape
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
SHAP value (impact on model output)
High
Low
SHAP impact magnitude

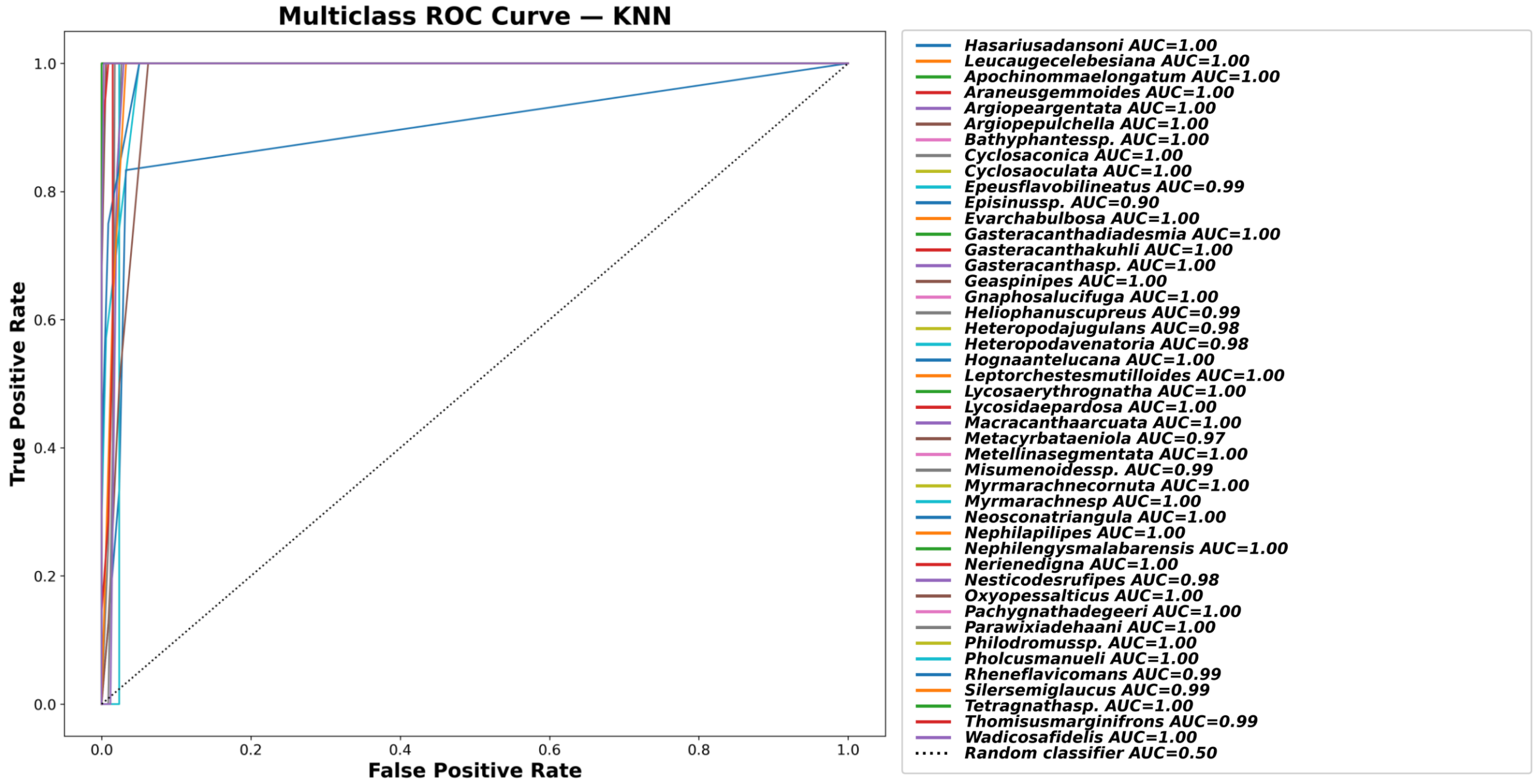

Global SHAP Beeswarm Plot — Logistic Regression
Predictor
Eye Spacing
Body length (mm)
Spinneret Type
Abdomen Shape
Abdominal Folium Pattern
Carapace Marking
Prosoma Shape
Chelicera Shape
Prosoma Length (mm)
Eye Pattern
Epigyne Structure
Epigyne Shape
Abdomen
Copulatory Duct Shape
Copulatory Opening Position
Prosoma width (mm)
Spinneret Arrangement
Retromarginal Teeth Count
Cheliceral Teeth Count (total per chelicera)
Spermathecae Shape
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
SHAP value (impact on model output)
High
Moderate
Low
Relative SHAP impact

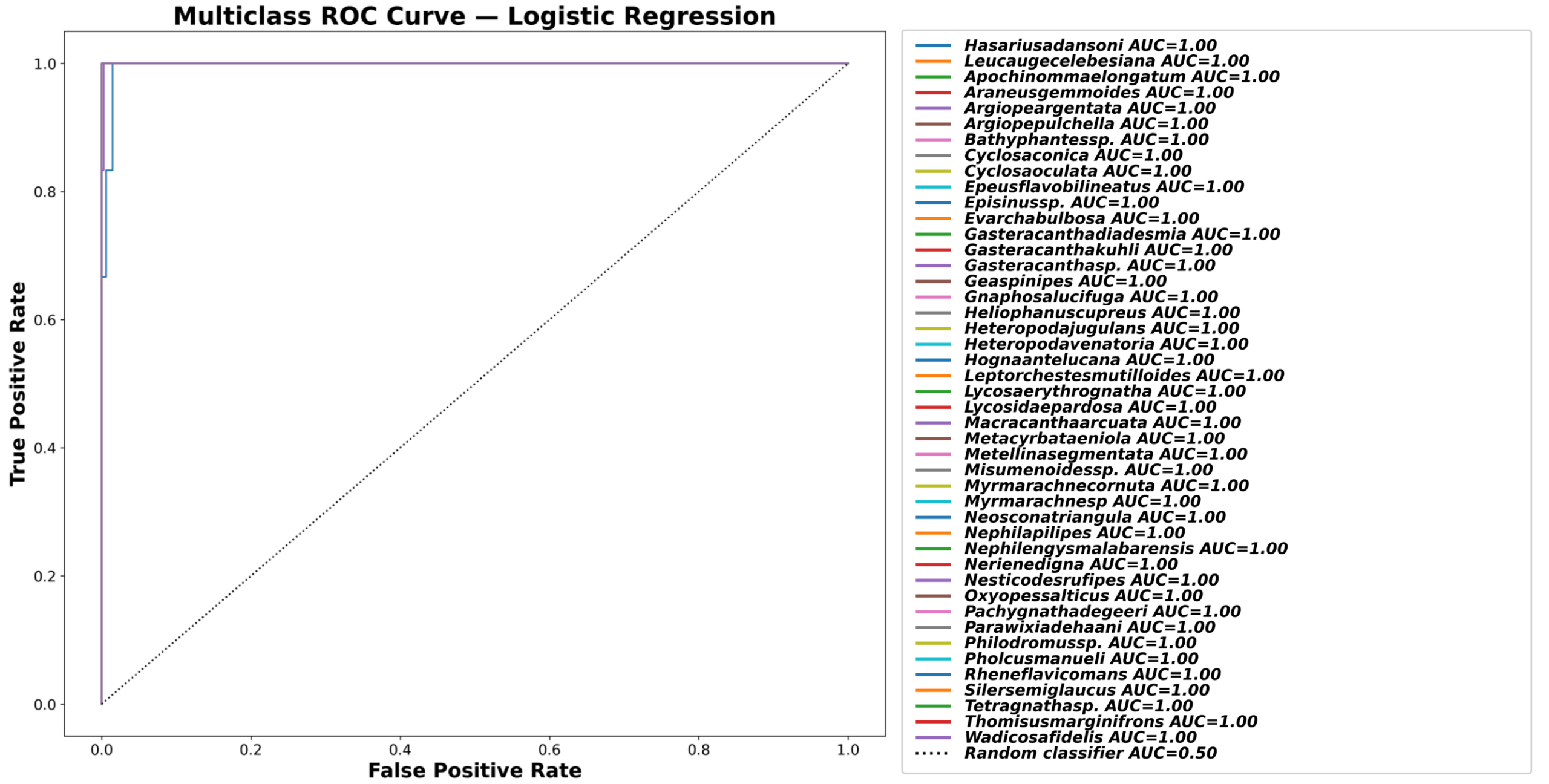

Global SHAP Beeswarm Plot — Random Forest
Predictor
Eye Spacing
Leg Formula
Abdomen
Trichobothria Pattern
Leg Spination Density
Eye Row Curvature
Anterior Eye Row
Spinneret Segmentation
Tibia Spines
Femur Spines
Metatarsus Spines
Body length (mm)
Spermathecae Shape
Copulatory Opening Position
Prosoma Length (mm)
Copulatory Duct Shape
Prosoma width (mm)
Retromarginal Teeth Count
Promarginal Teeth Count
Tarsal Claw Teeth
−0.05
0.00
0.05
0.10
0.15
SHAP value (impact on model output)
High
Moderate
Low
Relative SHAP impact

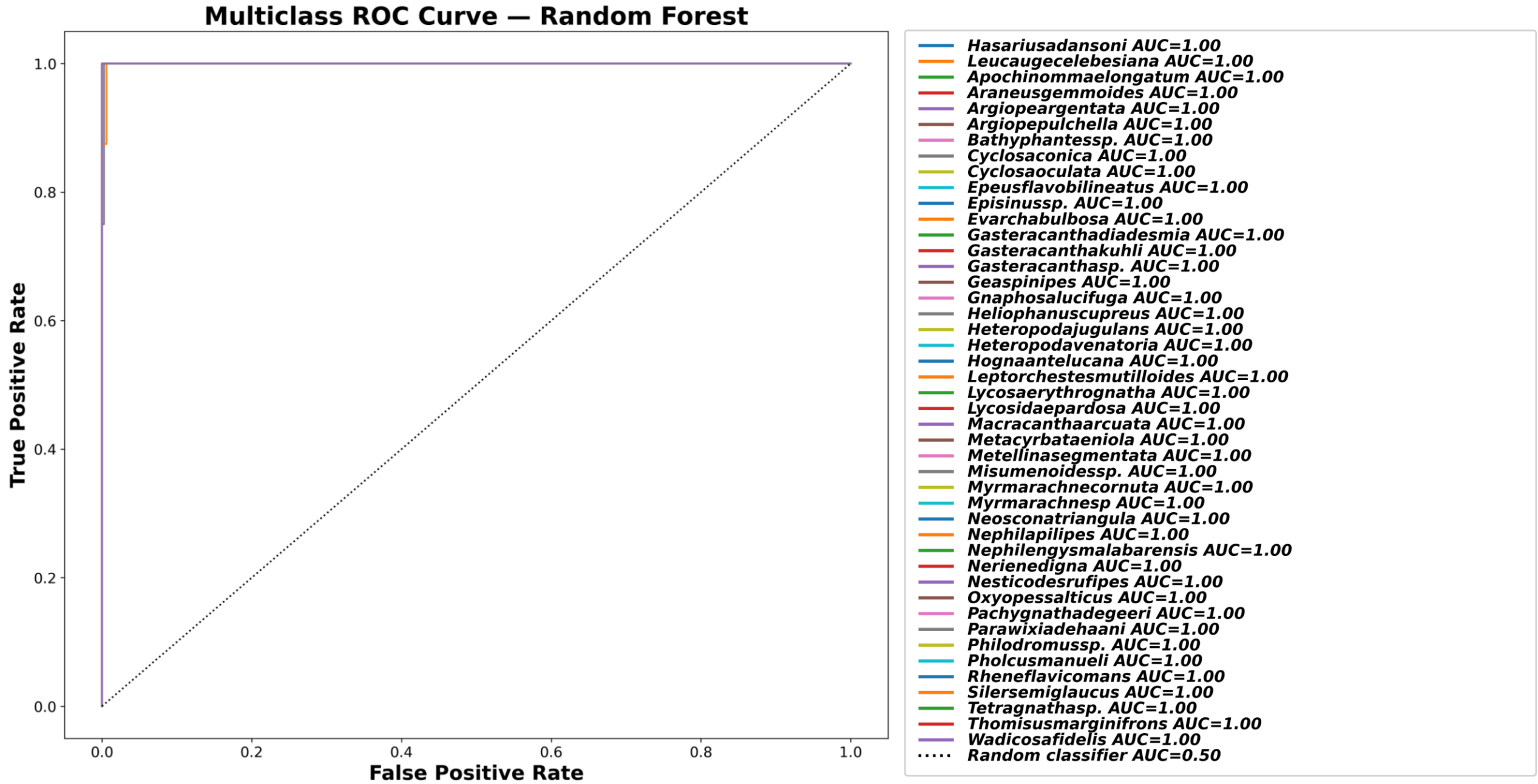

Global SHAP Beeswarm Plot — SVM
Predictor
Retromarginal Teeth Count
Eye Spacing
Promarginal Teeth Count
Prosoma Length (mm)
Prosoma width (mm)
Leg Formula
Eye Row Curvature
Cheliceral Teeth Count (total per chelicera)
Body length (mm)
Abdomen
Abdomen Shape
Leg Spination Density
Prosoma Shape
Trichobothria Pattern
Abdominal Folium Pattern
Carapace Marking
Eye Pattern
Chelicera Shape
Anterior Eye Row
Spinneret Type
−0.10
−0.05
0.00
0.05
0.10
0.15
SHAP value (impact on model output)
High
Moderate
Low
SHAP impact magnitude

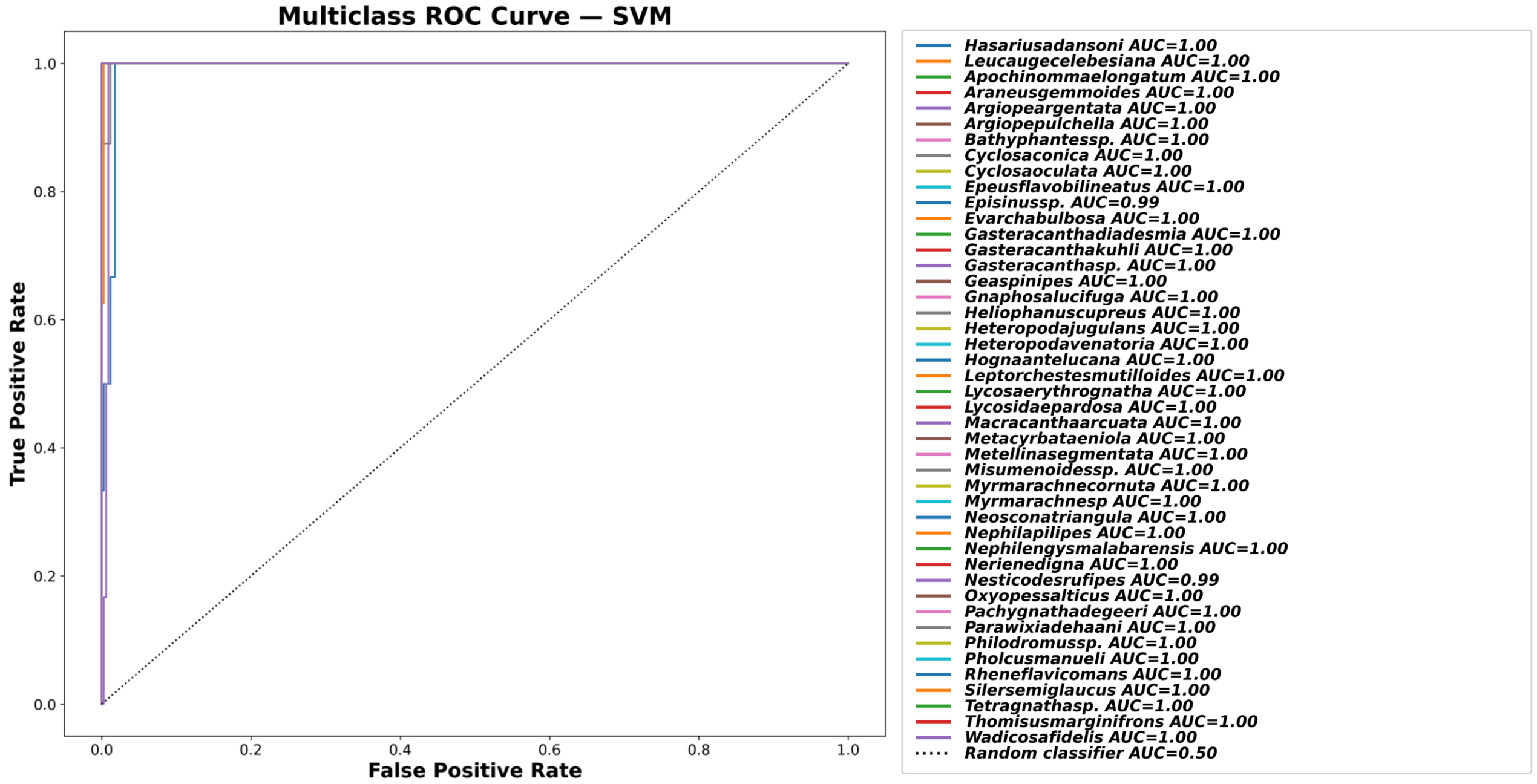

SHAP Feature Importance Across Five Machine-Learning Models
Taxonomic / Morphological Predictor
Prosoma width (mm) 0.0111 0.0021 0.0010 0.0020 0.0015
Eye Spacing 0.0012 0.0063 0.0017 0.0038 0.0020
Body length (mm) 0.0055 0.0026 0.0016 0.0028 0.0013
Prosoma Length (mm) 0.0047 0.0023 0.0011 0.0025 0.0017
Leg Formula 0.0025 0.0059 0.0009 0.0015 0.0014
Abdomen 0.0015 0.0057 0.0010 0.0022 0.0012
Trichobothria Pattern 0.0009 0.0057 0.0007 0.0013 0.0011
Eye Row Curvature 0.0001 0.0050 0.0008 0.0015 0.0013
Abdomen Shape 0.0000 0.0004 0.0035 0.0027 0.0012
Chelicera Shape 0.0000 0.0009 0.0032 0.0025 0.0011
Carapace Marking 0.0000 0.0007 0.0033 0.0026 0.0011
Prosoma Shape 0.0000 0.0005 0.0035 0.0026 0.0011
Promarginal Teeth Count 0.0013 0.0018 0.0011 0.0015 0.0018
Copulatory Duct Shape 0.0000 0.0022 0.0020 0.0022 0.0010
Retromarginal Teeth Count 0.0000 0.0020 0.0011 0.0018 0.0022
Decision Tree
Random Forest
KNN
Logistic Regression
SVM
Machine-Learning Model
Mean Absolute SHAP Value
0.010
0.008
0.006
0.004
0.002
0.000

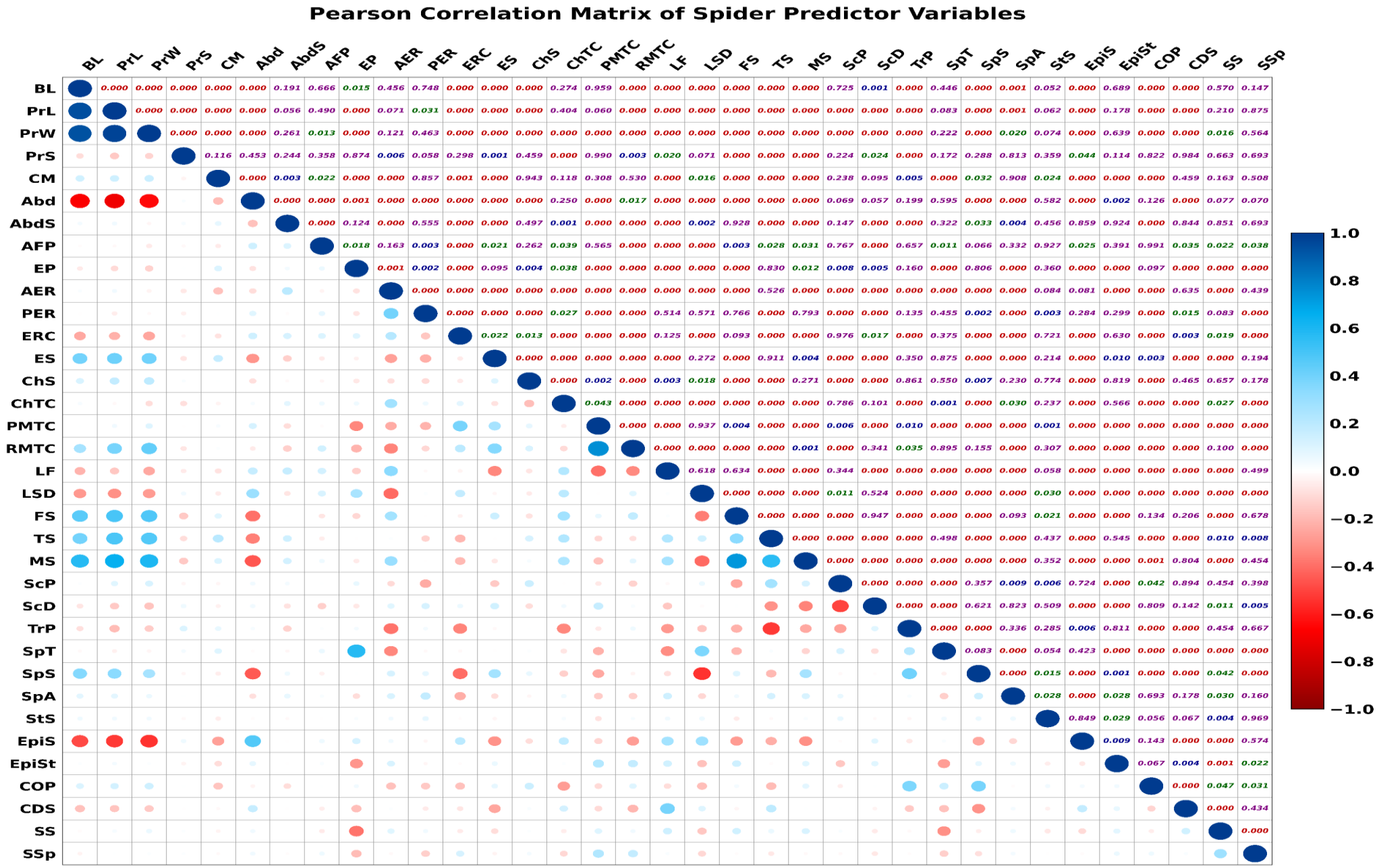

Multiclass ROC Curve — Decision Tree
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Hasariusadansoni AUC=1.00
Leucaugecelebesiana AUC=1.00
Apochinommaelongatum AUC=1.00
Araneusgemmoides AUC=1.00
Argiopeargentata AUC=1.00
Argiopepulchella AUC=1.00
Bathyphantessp. AUC=1.00
Cyclosaconica AUC=1.00
Cyclosaoculata AUC=1.00
Epeusflavobilineatus AUC=1.00
Episinussp. AUC=1.00
Evarchabulbosa AUC=1.00
Gasteracanthadiadesmia AUC=1.00
Gasteracanthakuhli AUC=1.00
Gasteracanthasp. AUC=1.00
Geaspinipes AUC=1.00
Gnaphosalucifuga AUC=1.00
Heliophanuscupreus AUC=1.00
Heteropodajugulans AUC=1.00
Heteropodavenatoria AUC=0.94
Hognaantelucana AUC=1.00
Leptorchestesmutilloides AUC=1.00
Lycosaerythrognatha AUC=1.00
Lycosidaepardosa AUC=1.00
Macracanthaarcuata AUC=1.00
Metacyrbataeniola AUC=1.00
Metellinasegmentata AUC=1.00
Misumenoidessp. AUC=1.00
Myrmarachnecornuta AUC=1.00
Myrmarachnesp AUC=1.00
Neosconatriangula AUC=1.00
Nephilapilipes AUC=1.00
Nephilengysmalabarensis AUC=1.00
Nerienedigna AUC=1.00
Nesticodesrufipes AUC=0.92
Oxyopessalticus AUC=1.00
Pachygnathadegeeri AUC=1.00
Parawixiadehaani AUC=1.00
Philodromussp. AUC=1.00
Pholcusmanueli AUC=1.00
Rheneflavicomans AUC=1.00
Silersemiglaucus AUC=1.00
Tetragnathasp. AUC=1.00
Thomisusmarginifrons AUC=1.00
Wadicosafidelis AUC=1.00
Random classifier AUC=0.50


Multiclass ROC Curve — KNN
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Hasariusadansoni AUC=1.00
Leucaugecelebesiana AUC=1.00
Apochinommaelongatum AUC=1.00
Araneusgemmoides AUC=1.00
Argiopeargentata AUC=1.00
Argiopepulchella AUC=1.00
Bathyphantessp. AUC=1.00
Cyclosaconica AUC=1.00
Cyclosaoculata AUC=1.00
Epeusflavobilineatus AUC=0.99
Episinussp. AUC=0.90
Evarchabulbosa AUC=1.00
Gasteracanthadiadesmia AUC=1.00
Gasteracanthakuhli AUC=1.00
Gasteracanthasp. AUC=1.00
Geaspinipes AUC=1.00
Gnaphosalucifuga AUC=1.00
Heliophanuscupreus AUC=0.99
Heteropodajugulans AUC=0.98
Heteropodavenatoria AUC=0.98
Hognaantelucana AUC=1.00
Leptorchestesmutilloides AUC=1.00
Lycosaerythrognatha AUC=1.00
Lycosidaepardosa AUC=1.00
Macracanthaarcuata AUC=1.00
Metacyrbataeniola AUC=0.97
Metellinasegmentata AUC=1.00
Misumenoidessp. AUC=0.99
Myrmarachnecornuta AUC=1.00
Myrmarachnesp AUC=1.00
Neosconatriangula AUC=1.00
Nephilapilipes AUC=1.00
Nephilengysmalabarensis AUC=1.00
Nerienedigna AUC=1.00
Nesticodesrufipes AUC=0.98
Oxyopessalticus AUC=1.00
Pachygnathadegeeri AUC=1.00
Parawixiadehaani AUC=1.00
Philodromussp. AUC=1.00
Pholcusmanueli AUC=1.00
Rheneflavicomans AUC=0.99
Silersemiglaucus AUC=0.99
Tetragnathasp. AUC=1.00
Thomisusmarginifrons AUC=0.99
Wadicosafidelis AUC=1.00
Random classifier AUC=0.50


Multiclass ROC Curve — Logistic Regression
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Hasariusadansoni AUC=1.00
Leucaugecelebesiana AUC=1.00
Apochinommaelongatum AUC=1.00
Araneusgemmoides AUC=1.00
Argiopeargentata AUC=1.00
Argiopepulchella AUC=1.00
Bathyphantessp. AUC=1.00
Cyclosaconica AUC=1.00
Cyclosaoculata AUC=1.00
Epeusflavobilineatus AUC=1.00
Episinussp. AUC=1.00
Evarchabulbosa AUC=1.00
Gasteracanthadiadesmia AUC=1.00
Gasteracanthakuhli AUC=1.00
Gasteracanthasp. AUC=1.00
Geaspinipes AUC=1.00
Gnaphosalucifuga AUC=1.00
Heliophanuscupreus AUC=1.00
Heteropodajugulans AUC=1.00
Heteropodavenatoria AUC=1.00
Hognaantelucana AUC=1.00
Leptorchestesmutilloides AUC=1.00
Lycosaerythrognatha AUC=1.00
Lycosidaepardosa AUC=1.00
Macracanthaarcuata AUC=1.00
Metacyrbataeniola AUC=1.00
Metellinasegmentata AUC=1.00
Misumenoidessp. AUC=1.00
Myrmarachnecornuta AUC=1.00
Myrmarachnesp AUC=1.00
Neosconatriangula AUC=1.00
Nephilapilipes AUC=1.00
Nephilengysmalabarensis AUC=1.00
Nerienedigna AUC=1.00
Nesticodesrufipes AUC=1.00
Oxyopessalticus AUC=1.00
Pachygnathadegeeri AUC=1.00
Parawixiadehaani AUC=1.00
Philodromussp. AUC=1.00
Pholcusmanueli AUC=1.00
Rheneflavicomans AUC=1.00
Silersemiglaucus AUC=1.00
Tetragnathasp. AUC=1.00
Thomisusmarginifrons AUC=1.00
Wadicosafidelis AUC=1.00
Random classifier AUC=0.50


Multiclass ROC Curve — Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Hasariusadansoni AUC=1.00
Leucaugecelebesiana AUC=1.00
Apochinommaelongatum AUC=1.00
Araneusgemmoides AUC=1.00
Argiopeargentata AUC=1.00
Argiopepulchella AUC=1.00
Bathyphantessp. AUC=1.00
Cyclosaconica AUC=1.00
Cyclosaoculata AUC=1.00
Epeusflavobilineatus AUC=1.00
Episinussp. AUC=1.00
Evarchabulbosa AUC=1.00
Gasteracanthadiadesmia AUC=1.00
Gasteracanthakuhli AUC=1.00
Gasteracanthasp. AUC=1.00
Geaspinipes AUC=1.00
Gnaphosalucifuga AUC=1.00
Heliophanuscupreus AUC=1.00
Heteropodajugulans AUC=1.00
Heteropodavenatoria AUC=1.00
Hognaantelucana AUC=1.00
Leptorchestesmutilloides AUC=1.00
Lycosaerythrognatha AUC=1.00
Lycosidaepardosa AUC=1.00
Macracanthaarcuata AUC=1.00
Metacyrbataeniola AUC=1.00
Metellinasegmentata AUC=1.00
Misumenoidessp. AUC=1.00
Myrmarachnecornuta AUC=1.00
Myrmarachnesp AUC=1.00
Neosconatriangula AUC=1.00
Nephilapilipes AUC=1.00
Nephilengysmalabarensis AUC=1.00
Nerienedigna AUC=1.00
Nesticodesrufipes AUC=1.00
Oxyopessalticus AUC=1.00
Pachygnathadegeeri AUC=1.00
Parawixiadehaani AUC=1.00
Philodromussp. AUC=1.00
Pholcusmanueli AUC=1.00
Rheneflavicomans AUC=1.00
Silersemiglaucus AUC=1.00
Tetragnathasp. AUC=1.00
Thomisusmarginifrons AUC=1.00
Wadicosafidelis AUC=1.00
Random classifier AUC=0.50


Multiclass ROC Curve — SVM
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Hasariusadansoni AUC=1.00
Leucaugecelebesiana AUC=1.00
Apochinommaelongatum AUC=1.00
Araneusgemmoides AUC=1.00
Argiopeargentata AUC=1.00
Argiopepulchella AUC=1.00
Bathyphantessp. AUC=1.00
Cyclosaconica AUC=1.00
Cyclosaoculata AUC=1.00
Epeusflavobilineatus AUC=1.00
Episinussp. AUC=0.99
Evarchabulbosa AUC=1.00
Gasteracanthadiadesmia AUC=1.00
Gasteracanthakuhli AUC=1.00
Gasteracanthasp. AUC=1.00
Geaspinipes AUC=1.00
Gnaphosalucifuga AUC=1.00
Heliophanuscupreus AUC=1.00
Heteropodajugulans AUC=1.00
Heteropodavenatoria AUC=1.00
Hognaantelucana AUC=1.00
Leptorchestesmutilloides AUC=1.00
Lycosaerythrognatha AUC=1.00
Lycosidaepardosa AUC=1.00
Macracanthaarcuata AUC=1.00
Metacyrbataeniola AUC=1.00
Metellinasegmentata AUC=1.00
Misumenoidessp. AUC=1.00
Myrmarachnecornuta AUC=1.00
Myrmarachnesp AUC=1.00
Neosconatriangula AUC=1.00
Nephilapilipes AUC=1.00
Nephilengysmalabarensis AUC=1.00
Nerienedigna AUC=1.00
Nesticodesrufipes AUC=0.99
Oxyopessalticus AUC=1.00
Pachygnathadegeeri AUC=1.00
Parawixiadehaani AUC=1.00
Philodromussp. AUC=1.00
Pholcusmanueli AUC=1.00
Rheneflavicomans AUC=1.00
Silersemiglaucus AUC=1.00
Tetragnathasp. AUC=1.00
Thomisusmarginifrons AUC=1.00
Wadicosafidelis AUC=1.00
Random classifier AUC=0.50



Upper triangle: p-values
- $p < 0.001$
- $p < 0.01$
- $p < 0.05$

Pearson Correlation Matrix of Spider Predictor Variables